

Yuwen Cui

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SUMMARY

Ph.D. Candidate in Computer Science specializing in AI security, cybersecurity, and privacy-preserving machine learning. Experienced designing security research prototypes, developing Python-based systems, and evaluating adversarial attacks and defenses for encrypted traffic. Strong background in machine learning, network security, large language models, and applied security research with multiple peer-reviewed publications. Passionate about building scalable detection, investigation, and response capabilities for modern cloud environments.

SKILLS

- **Languages:** Python, C/C++, Java, SQL, Bash
- **Security:** Network Security, AI Security, Adversarial ML, Traffic Analysis, Threat Modeling
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Large Language Models
- **Systems:** Linux, Git, Docker, Kubernetes
- **Research:** Experimental Design, Data Analysis, Privacy-Preserving ML

EXPERIENCE

University of South Florida | Tampa, FL

08/2021 - Current

Research Assistant

- Conducted research in cybersecurity and AI security, investigating novel attack and defense techniques to enhance modern computing systems and user privacy.
- Designed, implemented, and evaluated research prototypes, integrating programming, system development, and experimental analysis to advance research objectives.
- Performed literature reviews, analyzed experimental results, and drafted research manuscripts for peer-reviewed publication.

Teaching Assistant

- Led laboratory sessions and provided hands-on guidance with programming over 100 undergraduate students.
- Supported instruction in undergraduate courses on secure coding, enhancing student understanding of key concepts.
- Held weekly office hours to help students understand course material and debug programming projects.
- Evaluated assignments, projects, and exams with standardized rubrics, ensuring fair and consistent grading.

University of South Florida | Tampa, FL

05/2026 - 08/2026

NSF REU Mentor

- Mentored undergraduate researchers in NSF REU cybersecurity project on website fingerprinting and encrypted traffic analysis, guiding students in applying deep learning techniques for traffic classification, evaluating fingerprint attacks and defenses, and conducting experiments to address privacy challenges in network security.
- Guided students through academic and personal challenges effectively.
- Facilitated workshops on study skills and time management techniques.

University of North Texas | Denton, TX

01/2019 - 07/2021

Research Assistant

- Designed and implemented NoC architecture, evaluated system performance, and analyzed experimental data using state-of-the-art research methodologies.

- Developed research prototypes and reproducible experiments using Python, C/C++, Linux, and CUDA.
- Collaborated with faculty and graduate researchers to develop and submit projects to academic conferences, enhancing research visibility.

Teaching Assistant

- Supported instruction in undergraduate courses in Computer Organization and Architecture course with 100+ students.
- Conducted ARM programming labs and supported student learning.
- Conducted weekly office hours to clarify course material and assist students with programming project debugging.
- Graded assignments, projects, and exams using standardized rubrics.

EDUCATION

University of South Florida Tampa, FL Ph.D. in Computer Science	08/2021 – 12/2026
University of North Texas Denton, TX Doctor of Philosophy in Computer Science (Transferred)	01/2019 – 08/2021
China University of Petroleum Qingdao, China M.S. in Computer Science	08/2015 – 05/2018
Lanzhou University of Technology Lanzhou, China B.S. in Computer Science	09/2011 – 05/2015

PROJECTS

Privacy-Preserving Defenses Against Website Fingerprinting Attacks Developed privacy-preserving defense mechanisms against website fingerprinting attacks on encrypted traffic. Evaluated the robustness of existing defenses against adaptive machine learning-based attackers and explored techniques to improve privacy protection while maintaining network performance. • <i>Github:</i> https://github.com/WF-Attack-and-Defense/chameleon	09/2025 - current
Time-Series Large Language Models for Website Fingerprinting Attacks Developed a robust website fingerprinting framework that leverages time-series large language models (TSLMs) to improve resilience against model drift in encrypted traffic analysis. TSLM addresses the challenge of evolving network environments by learning long-term temporal traffic patterns and adapting to changes in website behaviors over time. The project explores the application of LLM-based sequence modeling for privacy analysis and demonstrates improved robustness against real-world traffic distribution shifts. • <i>Github:</i> https://github.com/WF-Attack-and-Defense/wf-tslm	09/2025 - 02/2026
Wi-Fi Fingerprint-Based Device Localization and Tracking in Public Wi-Fi Networks Developed a machine learning-based device localization and tracking system using Wi-Fi fingerprinting techniques with Received Signal Strength Indicator (RSSI) and Round-Trip Time (RTT) measurements. The project extracts wireless signal features from public Wi-Fi environments and applies machine learning models to estimate device locations and improve localization accuracy under dynamic network conditions. This work investigates the challenges of signal variability, wireless positioning, and privacy risks associated with device tracking in public Wi-Fi networks. • <i>Github:</i> https://github.com/RxStalker/RxStalker	05/2024 - 05/2025
Shared Resource Vulnerability Analysis in Serverless Computing Investigated security vulnerabilities in multi-tenant serverless computing environments by analyzing shared resource interference between co-resident serverless functions. Developed and evaluated a Shared Resource Entanglement (SRE) attack that demonstrates how attackers can infer sensitive information through memory access and resource usage patterns,	05/2022 - 05/2023

highlighting potential data leakage risks in Kubernetes-based serverless platforms and motivating stronger isolation mechanisms.

PUBLICATIONS

- Poster: Towards Model Drift Resistant Website Fingerprinting with Time-Series LLMs
- Shared Resource Entanglement Attacks against Serverless Computing
- A Multi-Agent Framework for High-Interaction Terminal Simulation
- A Comprehensive Survey of Website Fingerprinting Attacks and Defenses in Tor: Advances and Open Challenges
- A Low-Cost Conflict-Free NoC Architecture for Heterogeneous Multicore Systems

PATENTS

- A DNA Encryption-Based Hopping Covert Communication Method
- An Extended Sequence Covert Authentication Method Based on Address Knocking
- Node satellite communication method based on sequential decision route handover
- A non-blocking terminal information hopping method based on Web service
- FTP file reliable transmission method based on port hopping